Tathagata Debnath

Ph.D. Candidate, Computer Science (NMSU)

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Skills

Bioinformatics / Multi-Omics

LLM Fine-Tuning

PyTorch

RNA-seq / Proteomics Pipelines

Multi-GPU (H100 / A100 MIG)

RAG / FAISS / Prompt Eng.

Multi-Agent / LangGraph / MCTS

Classical ML (CNN, RNN, GAN)

Python / R / C++

Education

Ph.D., Computer Science
NMSU GPA 4.00 Expected 2026

M.S., CS (AI/ML)
NMSU GPA 4.00 Spring 2026

M.Tech, CSE (Gold Medal)
Tripura U. 9.83/10 2017

B.Tech, CSE
NIT Agartala 7.62/10 2015

Summary

PhD candidate at the intersection of **machine learning and biology**—building ML models for biological data, fine-tuning domain-specialized language models, and designing multi-omics pipelines. **Biology:** alternative splicing analysis (rMATS), proteomics pipelines, RNA-seq (Trinity, DESeq2), circular-genome clustering (IEEE TPAMI, TCBB). **ML:** end-to-end model training (PyTorch, QLoRA, DPO, multi-GPU). Built **BioLite Suite** (biology SLMs) and **CodeQ** (MCTS+DPO, **91% fix rate**). IEEE TPAMI / TCBB / Nature author with **81+ citations**.

Technical Stack

Bioinformatics: rMATS (splicing), DESeq2, Trinity, STAR/HISAT2, MaxQuant, Bioconductor, samtools, GO/pathway enrichment
ML/DL: PyTorch, LoRA/QLoRA, DPO/RLHF/SFT, Hugging Face (Transformers, TRL, PEFT), Flash Attention 2, bitsandbytes (NF4), vLLM, scikit-learn, XGBoost, SHAP
Agents: LangGraph, LangChain, Claude/OpenAI APIs, RAG, FAISS, MCTS
Infra: Multi-GPU (2×H100, NCCL), MIG A100, Docker, W&B, LangSmith, FastAPI
Languages: Python, R, C/C++, JavaScript, SQL, Bash

Selected Projects

BioLite Suite: Biology-Specialized SLMs — 2026 – Present
QLoRA, SFT, DPO, Llama 3.2, MIG A100

- Llama 3.2 (1B/3B) + QLoRA on MIG-partitioned A100 (∼4.8 GB) for **biology narrative generation**.
- Released 2 Hugging Face models, 1 curated biology dataset, full training pipeline.

NutriBot: Scientific Literature RAG — 2025 - 2026
FAISS, BM25, RRF, FastAPI, Claude API

- Hybrid FAISS + BM25 + RRF retrieval over **nutrition and biomedical literature**.
- Cut per-query LLM cost ∼**50%** via caching, threshold gating, and token budgeting.

Federated Diabetes Predictor — 2026
Python, Flower, PyTorch, FedAvg

- Privacy-preserving federated learning on **Diabetes Health Indicators**; evaluated convergence vs. centralized baselines.

CodeQ: Self-Improving Debugging Agent — 2026
PyTorch, DPO, MCTS, LoRA, 2×H100

- MCTS + DPO on Qwen2.5-Coder-7B; **84% fix rate on DebugBench** (base 43.9%); custom Fp32LogitsDPOTrainer for bf16 NaN fix.

Biology Research Experience

RA, Hansen Lab (Biology, NMSU) — 2023 – 2025
RNA-seq, rMATS, alternative splicing

- Analyzed radiation-induced **alternative splicing** in *Aedes aegypti*; identified **197 genes** across Hippo/Notch pathways (*Sci. Reports* 2025).

RA, Castillo Lab (Biology, NMSU) — 2021 – 2023
Proteomics, RNA-seq, Trinity, differential expression

- Built **proteomics/phospho-proteomics** and **RNA-seq** pipelines (Trinity, differential immune-response analysis).

Selected Publications

- Bendzus H. et al. (incl. **Debnath T.**). "Radiation-induced alternative splicing in *Aedes aegypti*." *Scientific Reports*, 2025.
- Chen Y., **Debnath T.**, Cai A., Song M. "Circular Silhouette and a Fast Algorithm." *IEEE TPAMI*, 45(11), 2023.
- **Debnath T.**, Song M. "Fast Optimal Circular Clustering on Round Genomes." *IEEE/ACM TCBB*, 18(6), 2021.